



Course Code: CSE 3107
Course Title: Communication Engineering

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Lecture # 10

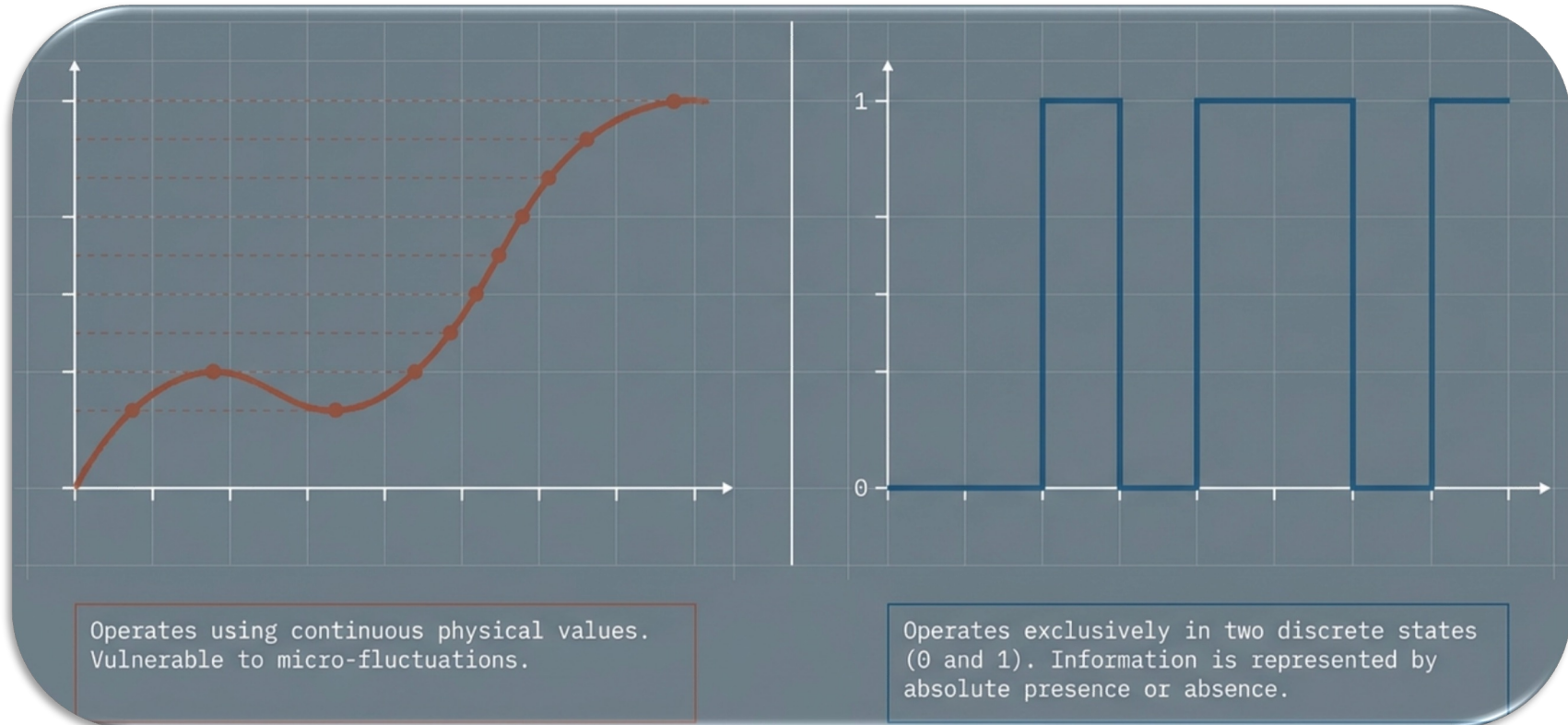
Digital Communication

Outline_____

- Basic Digital Communication System,
- A/synchronous Data Transmission, Synchronization
- Probability of Bit Error in Baseband Transmission,
- Matched Filter, and Eye Diagrams

Communication System Basic

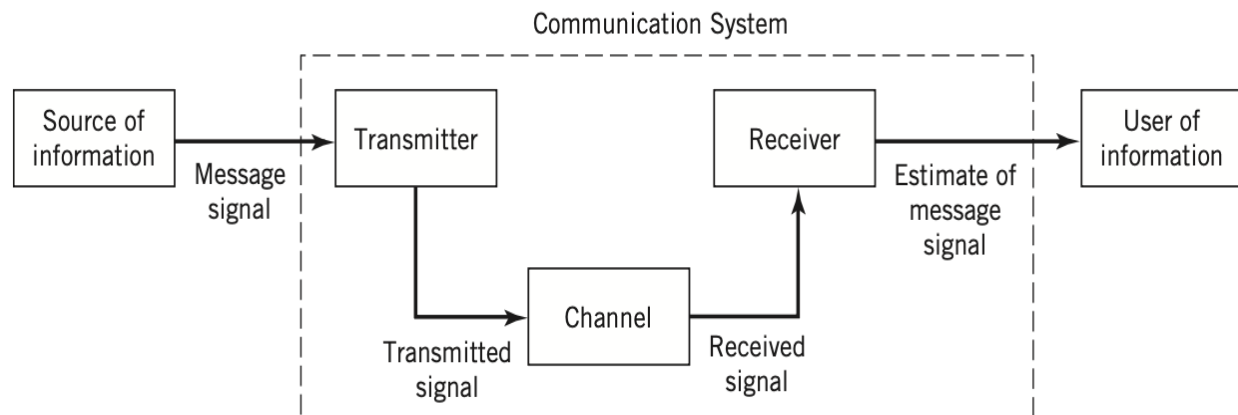
□ Analog vs Digital?



Communication System Basic

❑ Communication System?

- ✓ Communication system is defined as the setup that enables the transmission and reception of information (messages, signals, or data) from one point (transmitter/sender) to another point (receiver) through a communication channel.
- ✓ Irrespective of the form of communication process being considered, there are three basic elements to every communication system, namely, **transmitter**, **channel**, and **receiver**, as depicted in Figure 1.1.



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Figure 1.1 Elements of a communication system.

Communication System Basic

- **Communication System?**

✓ **A good communication system always considers these factors,**

- Collect information from the sender completely
- Transmit this information as much as possible in less time
- Deliver this information to the receiver without any error
- Transmit more information as much as possible.

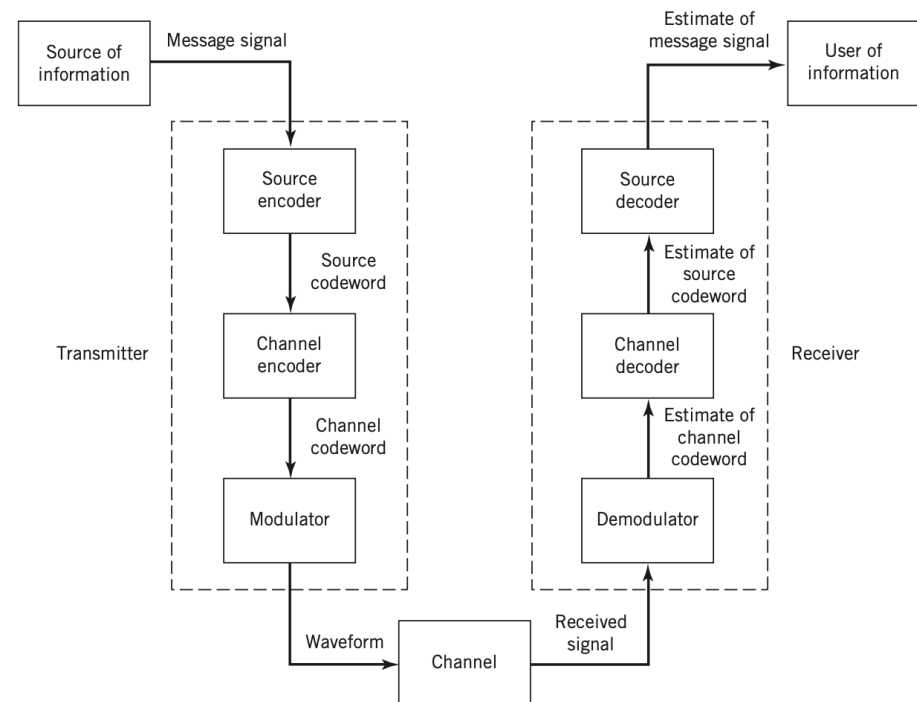


Figure 1.6 Block diagram of a digital communication system.

Communication System Basic

❑ Communication System?

- ✓ The functional block diagram illustrates the signal flow and the signal-processing steps through a typical digital communication system (DCS).

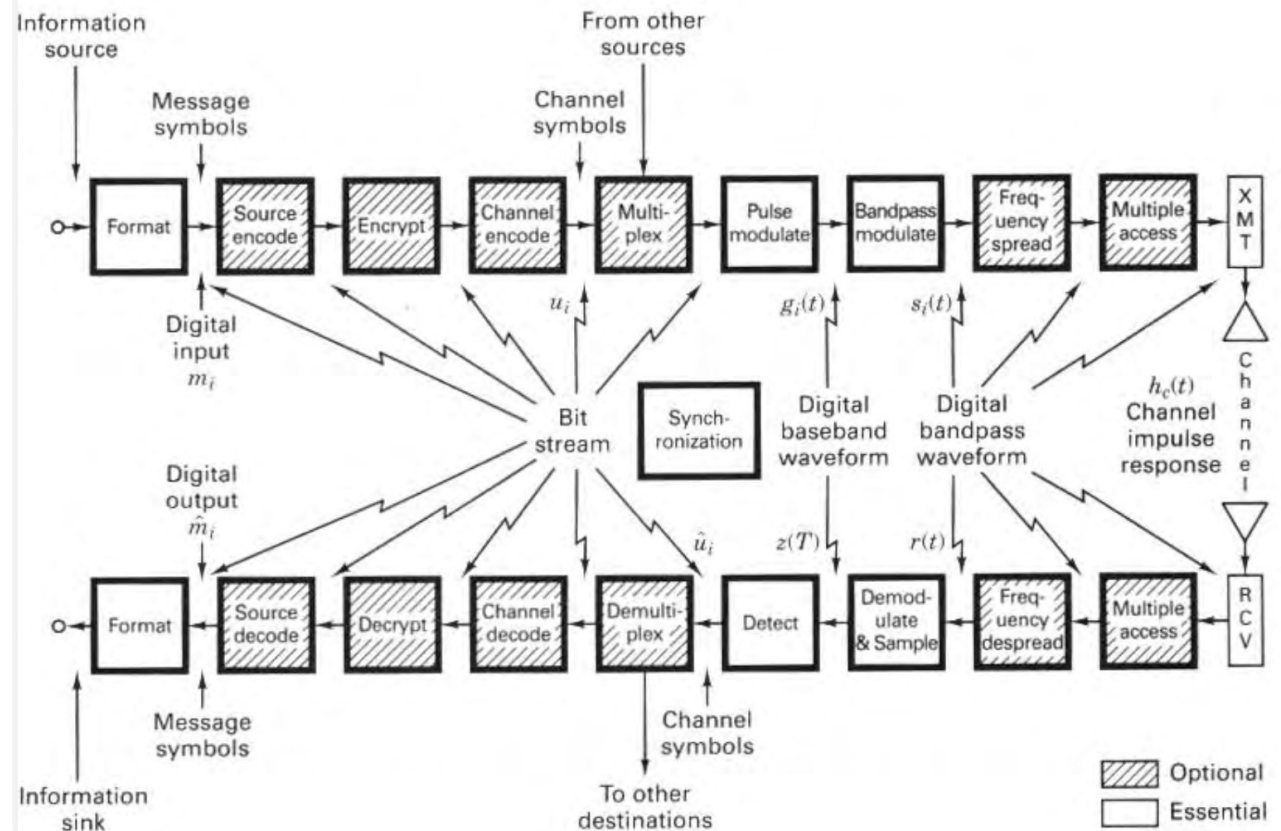


Figure 1.2 Block diagram of a typical digital communication system.

Communication System Basic

□ Communication System?

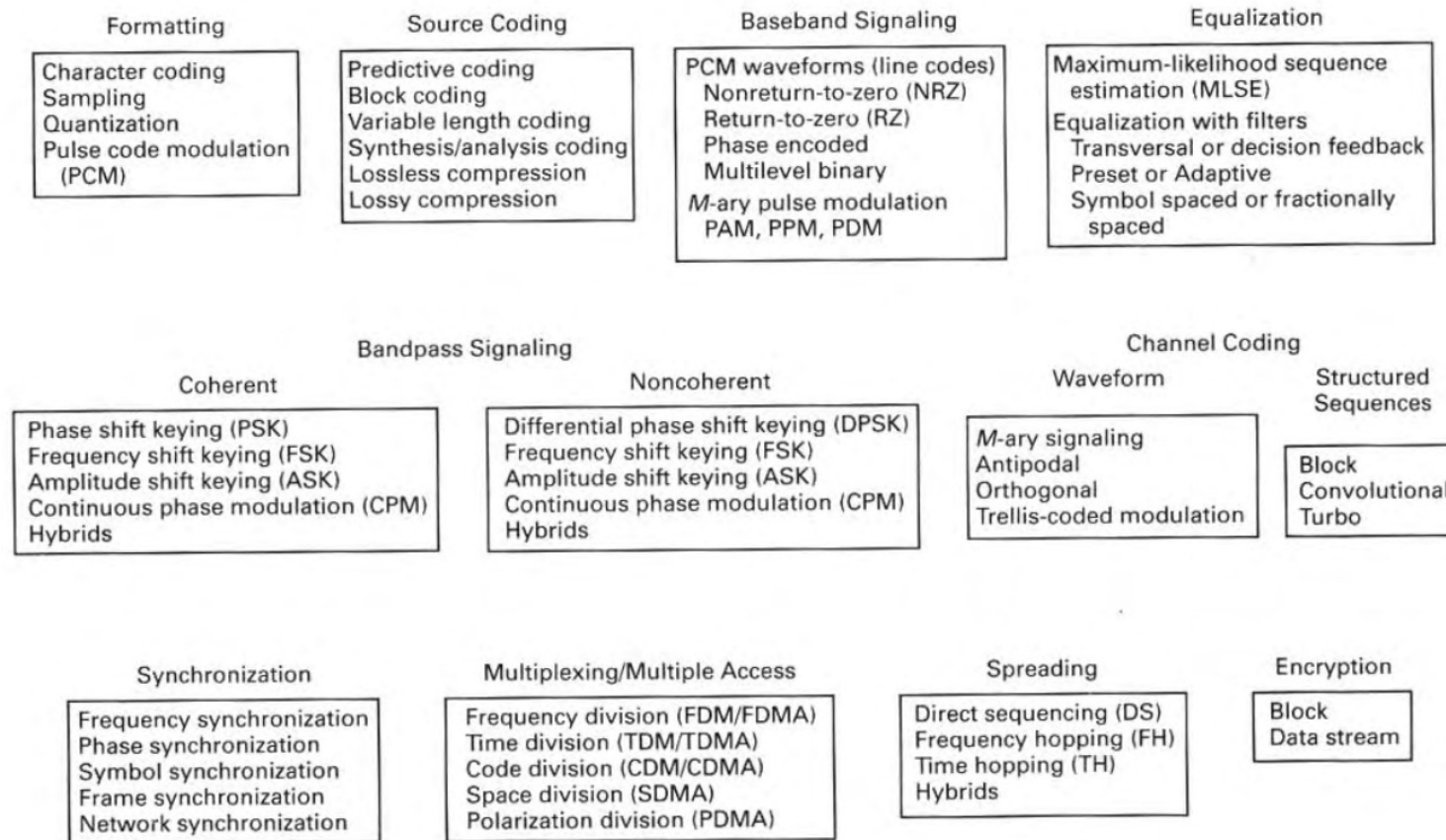


Figure 1.3 Basic digital communication transformations.

Communication System Basic

❑ Why Modern Systems Prefer Digital?

✓ Signal Regeneration

- Digital signals can be **regenerated (reborn)** using repeaters
- Prevents accumulation of noise over long distances

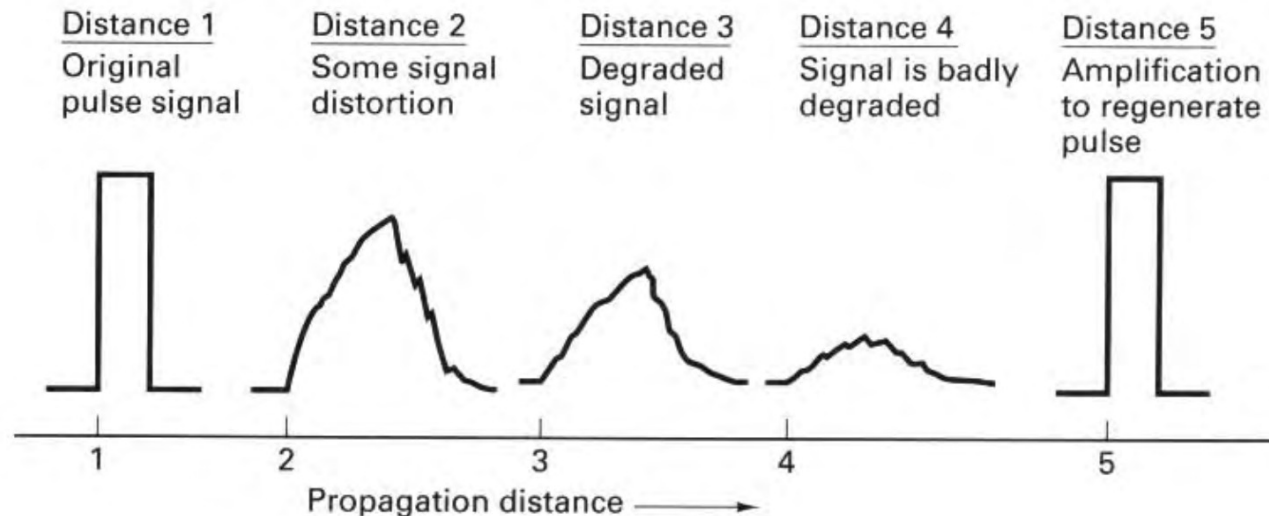


Figure 1.1 Pulse degradation and regeneration.

Communication System Basic

❑ Why Modern Systems Prefer Digital?

✓ Noise Immunity

- Operates in **two discrete states (0/1)**
- Small disturbances do not affect signal unless threshold is crossed

✓ High Signal Fidelity

- Supports **error detection and correction**
- Enables **very low error rates**, unlike analog

✓ Less Distortion

- Digital signals resist distortion better
- Analog distortion is **permanent and cannot be removed by amplification**

Communication System Basic

❑ Why Modern Systems Prefer Digital?

- ✓ **Higher Reliability & Lower Cost**
 - Digital circuits are **more reliable**
 - Easier and cheaper to manufacture (LSI, microprocessors)
- ✓ **Flexibility**
 - Easily implemented with **programmable hardware/software**
 - Supports a wide range of applications
- ✓ **Efficient Multiplexing**
 - **Time-Division Multiplexing (TDM)** is simpler than analog FDM

Communication System Basic

❑ Why Modern Systems Prefer Digital?

- ✓ **Uniform Signal Handling**
 - All data types treated equally → *“a bit is a bit”*
- ✓ **Packet-Based Transmission**
 - Data can be sent in **packets**, enabling efficient switching
- ✓ **Security & Processing**
 - Supports **encryption, privacy, and interference protection**
- ✓ **Better for Modern Systems**
 - Ideal for **computer-to-computer communication** and digital devices

Communication System Basic

❑ Costs of Digital Communication System Benefits ?

1. High Signal Processing Requirement: Digital systems are **more signal-processing intensive** than analog systems.

- ✓ They require complex operations like encoding, decoding, compression, and error correction.
- ✓ This increases system complexity, cost, and power consumption.

2. Need for Extensive Synchronization: Digital communication requires **precise synchronization at multiple levels** (e.g., timing, frame, bit level).

- ✓ A large portion of system resources is dedicated to maintaining synchronization.
- ✓ In contrast, analog systems achieve synchronization more easily and with less effort.

Communication System Basic

❑ Costs of Digital Communication System Benefits ?

3. Nongraceful Degradation (Sudden Performance Drop): Digital systems suffer from **nongraceful degradation**:

- ✓ When the **signal-to-noise ratio (SNR)** falls below a certain threshold, performance drops abruptly.
- ✓ The output can suddenly change from **clear and accurate to highly distorted or unusable**.

4. Less Smooth Performance Compared to Analog

- ✓ Analog systems degrade **gradually (gracefully)** as noise increases.
- ✓ Digital systems, however, maintain good quality up to a point, then **fail sharply**, which can be undesirable in some applications.

Communication System Basic

□ Digital versus Analog Performance Criteria:

- ✓ A principal difference between analog and digital communication systems has to do with the way in which we evaluate their performance.
 - The figure of merit for the performance of analog communication systems is a fidelity criterion, such as **signal-to-noise ratio, percent distortion, or expected mean-square error** between the transmitted and received waveforms.

By contrast,

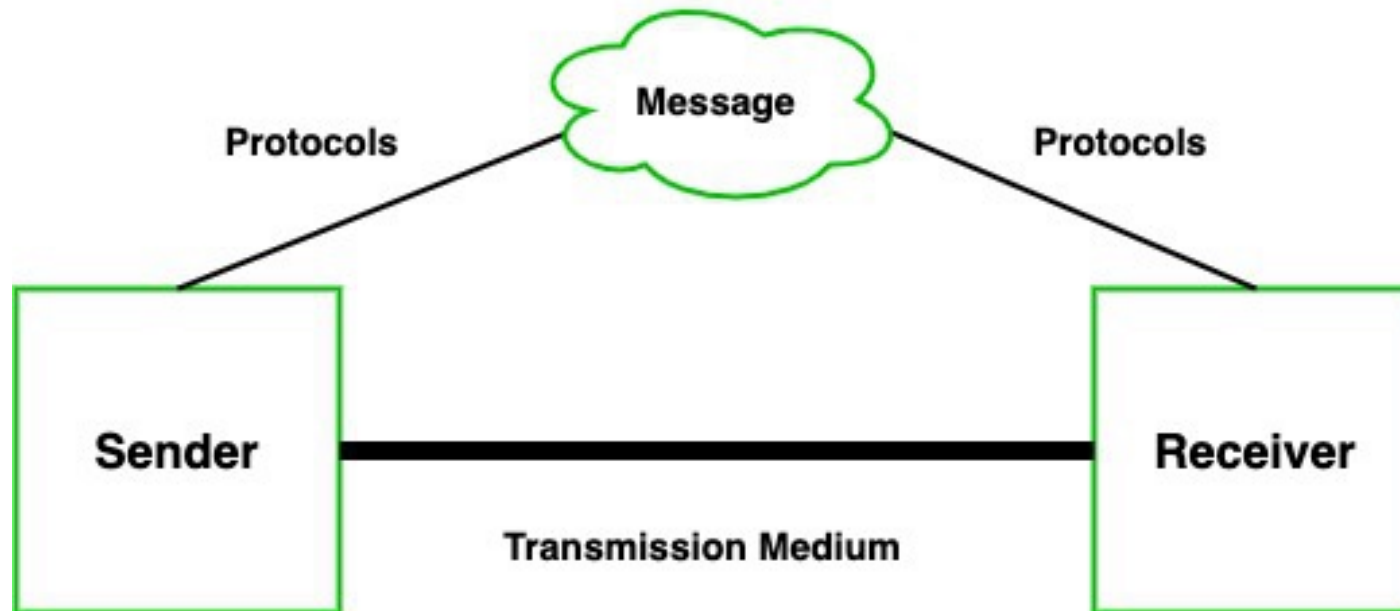
- A figure of merit for digital communication systems is the probability of incorrectly detecting a digit, or **the probability of error (P_e)**.

Data Transmission & its Types:

- ❑ **Data Transmission:** It refers to the transfer of data in the form of digital or analog signals from one device to another.
- ❑ Data transmission involves at least two or more digital devices communicating over a network, and requires a few key components:
 - ✓ **Sender:** The device that starts the transmission of data.
 - ✓ **Receiver:** The device that receives the data sent by the sender.
 - ✓ **Message or data:** text, images, audio, video, or any other form of content.
 - ✓ **Medium:** The physical path or channel through which data is transmitted, such as an optical cable or wireless transmission.
 - ✓ **Protocol:** A set of rules governing the format, timing, and sequencing of data transmission.

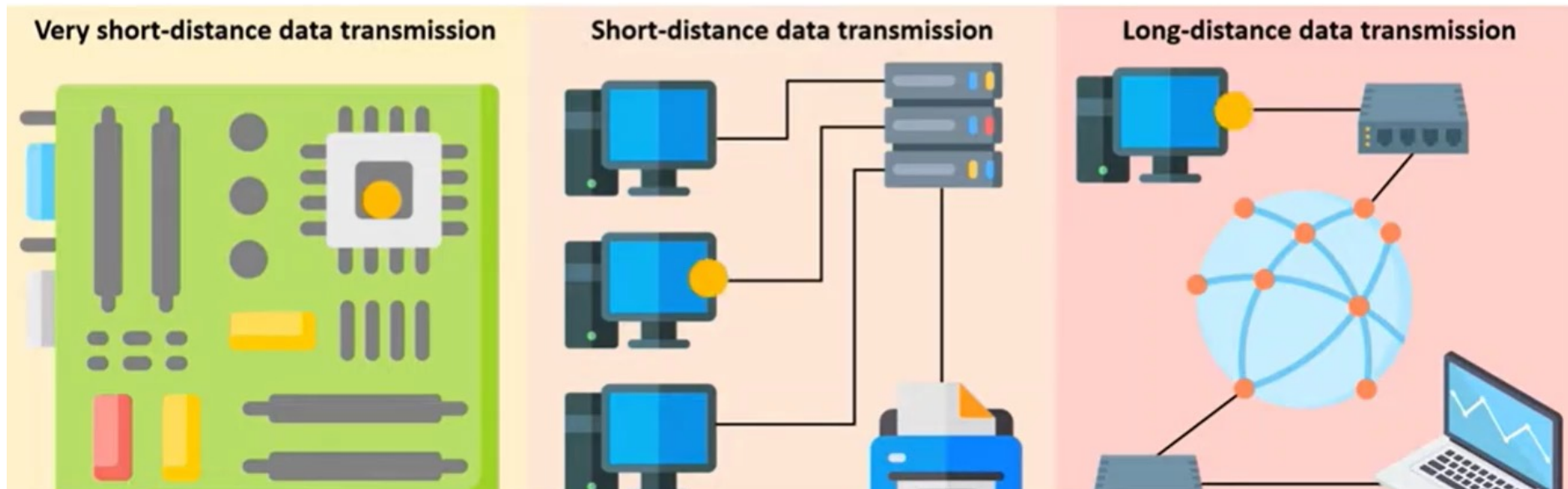
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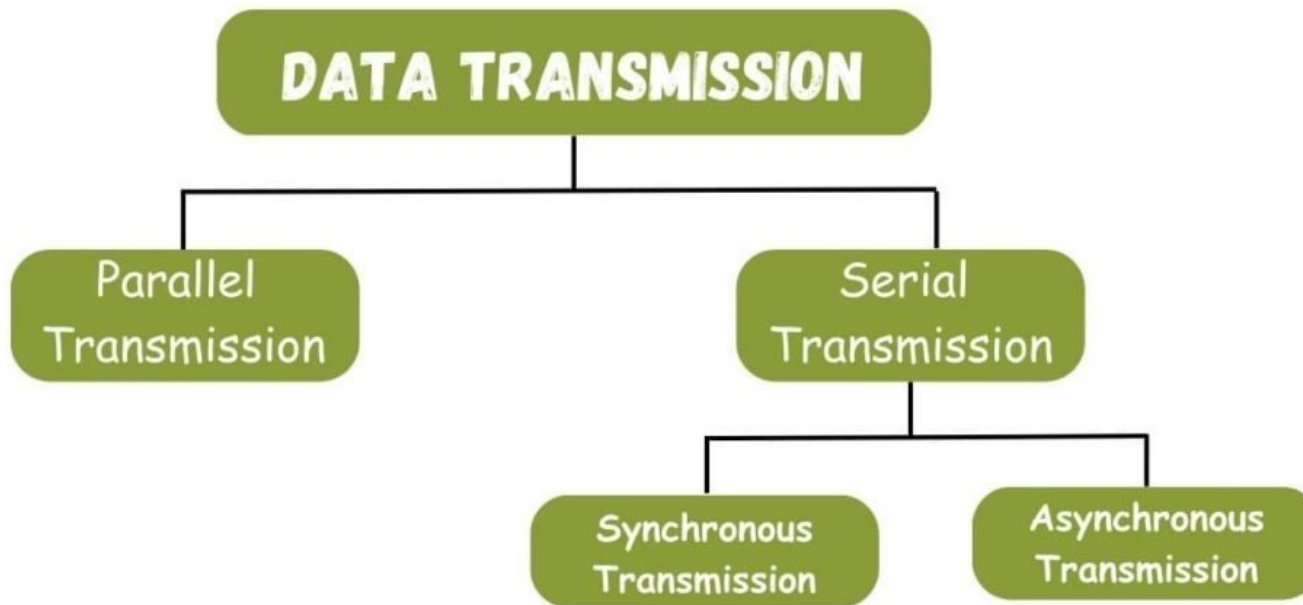
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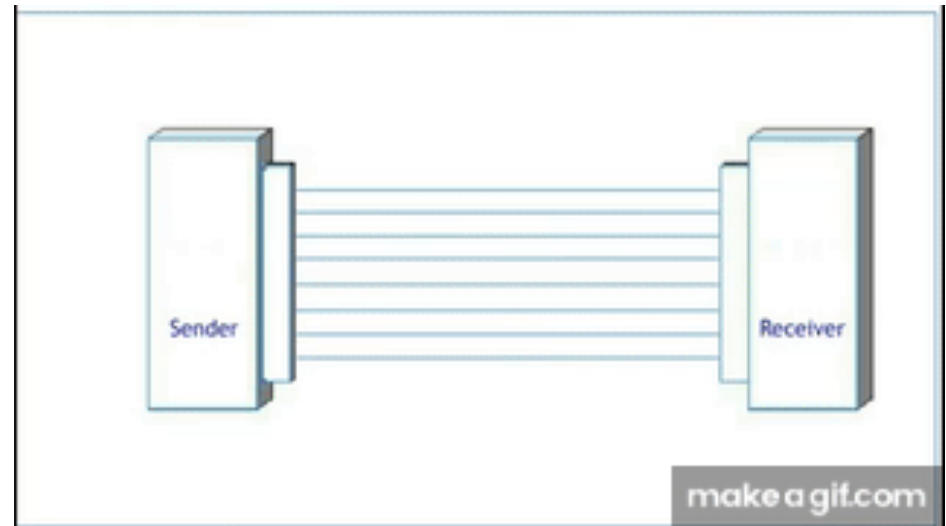
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Data Transmission & its Types:

❑ Parallel Transmission:

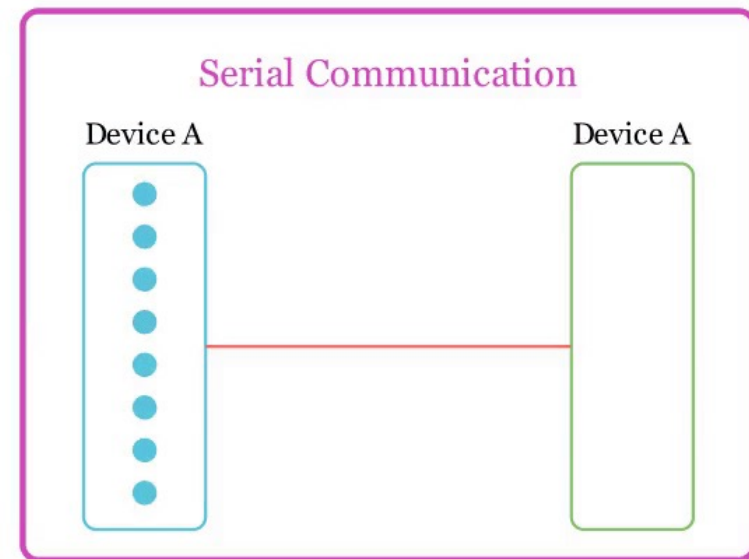
- ✓ This method transmits multiple bits (8, 16, 32, 64 bits) simultaneously across multiple channels.
- ✓ While it allows for faster data transfer rates, it requires **more bandwidth** and is typically used for **short distances**, such as within a computer.
- ✓ Example: USB port, Printers



Data Transmission & its Types:

❑ Serial Transmission:

- ✓ In this method, data is sent one bit at a time over a single channel.
- ✓ This approach is simpler and reduces the risk of signal interference, making it suitable for long-distance communication.
- ✓ Examples: Mouse, keyboard, card reader.



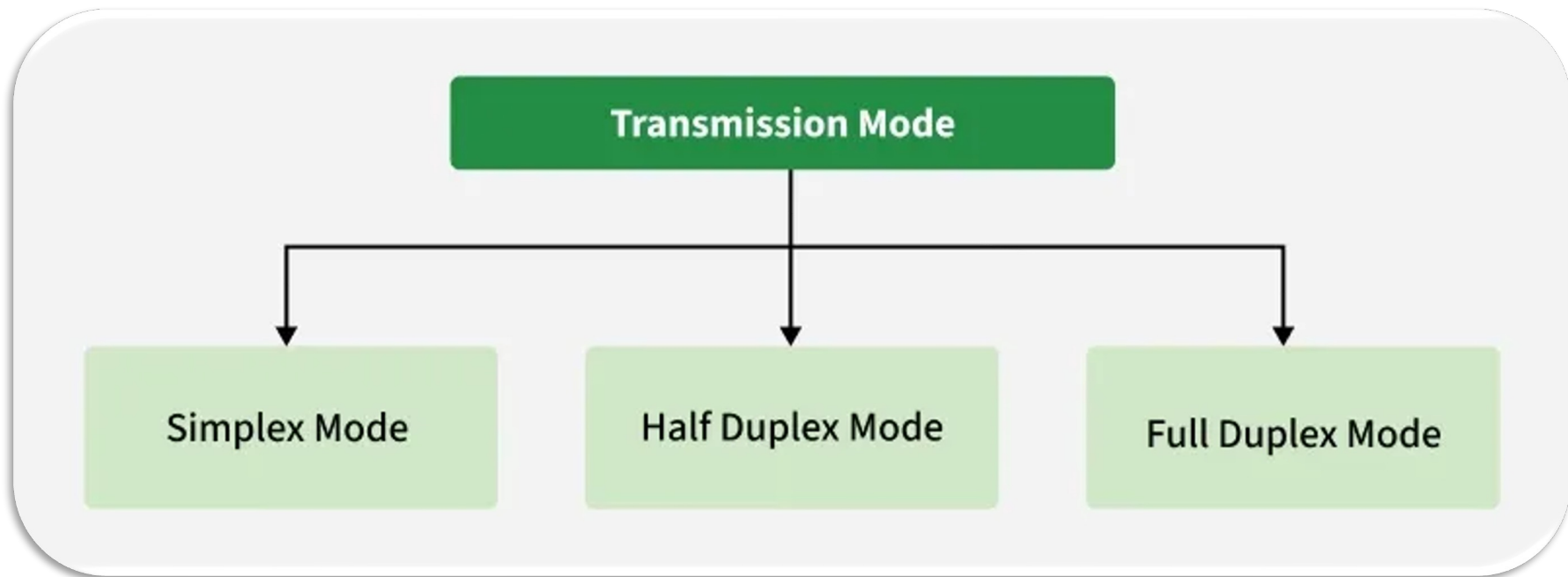
Data Transmission & its Types:

❑ Serial vs Parallel Data Transmission:

Feature	Serial Transmission	Parallel Transmission
Data Transfer Method	1 bit at a time	Multiple bits at once
Speed	Slower per clock cycle, but very high with modern technology	Faster per cycle, limited by distance/interference
Number of Wires	Few (often 1–2)	Many (e.g., 8 or more)
Distance	Better for long distances	Best for short distances
Cost & Complexity	Cheaper and simpler	More expensive and complex
Reliability	More reliable over long distances	Prone to timing errors and crosstalk
Example	Commonly used with the USB (universal Serial Bus) interface	Commonly used with printer cables and other internal device connection

Data Transmission & its Types:

- ❑ **Data Transmission (Communication) Modes:** Describe how data is exchanged between two devices over a communication channel.



Data Transmission & its Types:

❑ Simplex Mode:

- ✓ Data flows in only one direction
- ✓ One-way only
- ✓ Example: Keyboard → Computer

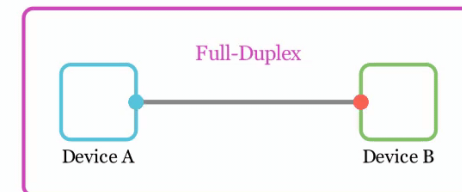
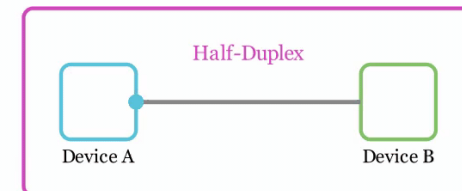
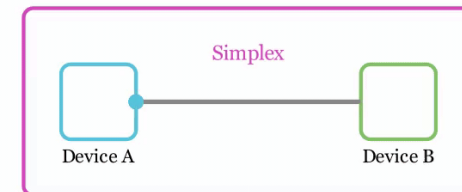
❑ Half-Duplex Mode:

- ✓ Data flows in both directions, but not at the same time
- ✓ Two-way (one at a time)
- ✓ Example: Walkie-talkie

❑ Full-Duplex Mode:

- ✓ Data flows in both directions simultaneously
- ✓ Two-way (simultaneous)
- ✓ Example: Phone call

Communication Modes



Thank You All